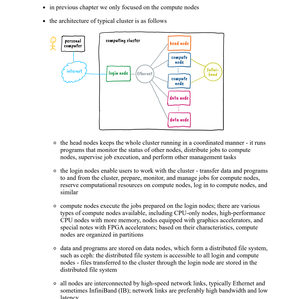
Cluster

Hardware

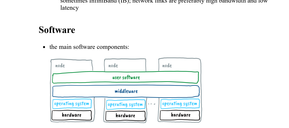
- in previous chapter we only focused on the compute nodes
- the architecture of typical cluster is as follows



  - the head nodes keeps the whole cluster running in a coordinated manner - it runs programs that monitor the status of other nodes, distribute jobs to compute nodes, supervise job execution, and perform other management tasks
  - the login nodes enable users to work with the cluster - transfer data and programs to and from the cluster, prepare, monitor, and manage jobs for compute nodes, reserve computational resources on compute nodes, log in to compute nodes, and similar
  - compute nodes execute the jobs prepared on the login nodes; there are various types of compute nodes available, including CPU-only nodes, high-performance CPU nodes with more memory, nodes equipped with graphics accelerators, and special nodes with FPGA accelerators; based on their characteristics, compute nodes are organized in partitions
  - data and programs are stored on data nodes, which form a distributed file system, such as ceph: the distributed file system is accessible to all login and compute nodes - files transferred to the cluster through the login node are stored in the distributed file system
  - all nodes are interconnected by high-speed network links, typically Ethernet and sometimes Infiniband (IB); network links are preferably high bandwidth and low latency

Software

- the main software components:



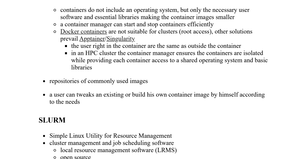
  - operating system
    - performs basic tasks such as memory management, processor management, device control, file system management, security functions, system operation control, resource usage monitoring, and error detection
    - preferably open source (AlmaLinux)
  - middleware
    - connects the operating system and user applications within the cluster
    - it ensures the coordinated operation of multiple nodes, enables centralized node management, handles user authentication, controls job execution (user applications) on the compute nodes
    - preferably open-source software:
      - automation through Foreman, Puppet, Ansible
      - administration and metrics: ElasticSearch, Syslog, Icinga, Nagios, IPMI, monitoring and visualization: Prometheus, Grafana
      - storage and data handling: dCache, Ceph, Rucio, iRODS, BeeGFS
      - job scheduling: SLURM
      - job submission system: ARC
      - container and virtualization support: Apptainer/Singularity, Proxmox, OpenStack
  - user software
    - with user software, users perform desired functions
    - user software is the reason why users use clusters
    - only user software adapted for the Linux operating system can be used in clusters
    - the user software can be installed on clusters in various ways:
      - an administrator installs it directly on the nodes (GNU toolchain)
      - an administrator prepares environmental modules
      - an administrator prepares containers for general use
      - a user installs it in their user space (home directory)
      - a user prepares a container in his user space

Environmental Modules

- mechanism for modifying system settings (environment variables)
- environmental module file contains the necessary information to configure system settings for specific software
- they are prepared and updated by the cluster administrator
- they simplify maintenance, avoid issues related to using different library versions
- users can load and unload available modules during their work
- when loading an environmental module, environment variables are adjusted for executing the selected user software, for example, PATH specifies the directories where the operating system searches for executables

Virtualization and Containers

- when directly installing user applications above the operating system, compatibility issues can arise, most commonly due to incompatible library versions
- virtualizing the nodes is an elegant solution that ensures the coexistence of diverse user applications and, therefore, more straightforward system management
- the system's performance is slightly reduced due to virtualization, but it has increased robustness and ease of maintenance
- hardware virtualization (virtual machines) and operating system virtualization (containers)
- for clusters, container-based virtualization is more suitable



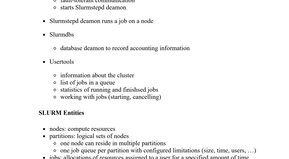
  - containers do not include an operating system, but only the necessary user software and essential libraries making the container image smaller
  - a container manager can start and stop containers efficiently
  - Docker containers are not suitable for clusters (root access), other solutions prevail: Apptainer/Singularity
    - the user right in the container are the same as outside the container
    - in an HPC cluster the container manager ensures the containers are isolated while providing each container access to a shared operating system and basic libraries
- repositories of commonly used images
- a user can tweaks an existing or build his own container image by himself according to the needs

SLURM

- Simple Linux Utility for Resource Management
- cluster management and job scheduling software
  - local resource management software (LRMS)
  - open source
  - fault tolerant
  - highly scalable
  - a lot of plugins (accounting, network, MPI)
- key functions
  - allocates access to resources (compute nodes) to users
  - framework for starting, executing and monitoring work
  - arbitrates contention for resources by managing a queue of pending work
- resource manager
  - needed in a parallel computer to execute parallel jobs
  - allocates resources within a cluster
    - nodes
      - sockets, cores, threads
      - memory
      - interconnect
      - features (GPUs, bigmem)
    - licenses
  - manages jobs through a queue
    - every job is sent to queue where it waits for available resources
    - complex scheduling algorithms
    - resource time-limit

SLURM architecture

- image from the SLURM website



- SlurmctId daemon on management node
  - monitors resources
  - manages job queues
  - allocates resources
  - optional fail-over twin
- Slurmd deamon runs on each node
  - similar to remote shell
  - waits for work, executes work, reports status
  - hierarchical concept
  - fault-tolerant communication
  - starts Slurmstepd deamon
- Slurmstepd deamon runs a job on a node
- Slurmdbs
  - database deamon to record accounting information
- User tools
  - information about the cluster
  - list of jobs in a queue
  - statistics of running and finished jobs
  - working with jobs (starting, cancelling)

SLURM Entities

- nodes: compute resources
- partitions: logical sets of nodes
  - one node can reside in multiple partitions
  - one job queue per partition with configured limitations (size, time, users, …)
- jobs: allocations of resources assigned to a user for a specified amount of time
- job steps: sets of (possibly parallel) tasks within a job

SLURM jobs

- jobs are allocated nodes within a partition (according to the priorities) until the resources (nodes, processors, memory, etc.) within that partition are exhausted
- once a job is assigned a set of nodes, the user is able to initiate parallel work in the form of job steps
  - a single job step may be started that utilizes all nodes allocated to the job
  - several job steps may independently use a portion of the allocation
- multiple job steps can be simultaneously submitted as they queued until there are available resources within the job's allocation
- a job lifecycle

  - job states:
    - pending (PD), running (R), completing (CG), completed (CD),
    - failed (F), timeouted (TO), suspended (S), revoked (RV), cancelled (CA),
    - node failure (NF), special exit (SE), configuring (CF)

Important commands

- sinfo: information on node state
  - down, draining, drained, failing, fail, reboot, maintenance, power, …
  - idle, allocated, mixed, completing, reserved
    - idle: all cores are available on the compute node
    - mix: at least one core is available on the compute node
    - alloc: all cores on the compute node are assigned to jobs
    - *: node is not responding, will not take new workload
- squeue: status of jobs and job-steps
- srun: create job allocation and launch job steps
  - blocks the shell
- sbatch: submit script for later execution (batch mode)
  - does not block the shell
  - many possibilities for job specification
    - job dependence (flag –dependency)
    - run many jobs with different params
- salloc: starts the shell on the first node that corresponds to the given requirements
  - interactive work
  - resources are specified in the same way as for srun/sbatch
  - exit to release resources
- scontrol: job control (hold, release, show nodes, …), system config
- sstat: statistics of active job
- sacct: statistics of active and finished jobs

SLURM examples

- basic cluster info

```
sinfo
sinfo --Node --long
sinfo --reservation
scontrol show node wn181
scontrol show partition
```

- running four tasks on the same and on two nodes

```
srun --ntasks=4 hostname
srun --ntasks=4 --nodes=2 hostname
```

- running a batch job (from files folder) and checking the queue status

```
sbatch hn.sh
squeue --me
```

- interactive work

```
salloc --ntasks=2
srun hostname
exit
```

- using modules

```
module spider
echo $PATH
srun ffmpeg -version
module load ffmpeg
module list
echo $PATH
srun ffmpeg -version
```

```
srun ffmpeg -y -i llama.mp4 llama.mp3
sacct --jobs=<job id> --s
      format=job,cputime,elapsed,AveRSS,AveVMSize,MaxRSS,MaxVMSize
```

- using containers

```
apptainer pull docker://jrottenberg/ffmpeg:alpine
srun apptainer exec ffmpeg_alpine.sif ffmpeg -version
srun apptainer exec ffmpeg_alpine.sif ffmpeg -y -i llama.mp4 llama.mp3
```